

Subnet Overview

Network Name	Subnet	Subnet Mask	CIDR	IP Range	DHCP	Purpose
Internal Network	192.168.1.0	255.255.255.0	/24	192.168.1.1 - 254	Yes	All office devices
Guest Network	192.168.2.0	255.255.255.0	/24	192.168.2.1 - 254	Yes	Isolated guest Wi-Fi

Static IP Assignments

Device	Hostname	IP Address	Subnet	Purpose
Router / Firewall	ubuntu-router	192.168.1.1	Internal	Default gateway, DHCP server, VPN
Network Switch	ubuntu-switch	192.168.1.2	Internal	Management access
NAS Device	ubuntu-nas	192.168.1.10	Internal	File storage and backups
Local Server	ubuntu-server	192.168.1.11	Internal	Development, database, test environment
WAP 1 (Office)	ubuntu-wap-office	192.168.1.20	Internal	Office Wi-Fi for laptops
WAP 2 (Guest)	ubuntu-wap-guest	192.168.1.21	Internal	Guest Wi-Fi (isolated)
Printer 1 (if added later)	ubuntu-printer-1	192.168.1.30	Internal	Shared printing (Finance wing)
Printer 2 (if added later)	ubuntu-printer-2	192.168.1.31	Internal	Shared printing (Dev wing)

Dynamic IP Ranges

Device Type	IP Range	Subnet	Number of Addresses	Notes
Desktop PCs (20)	192.168.1.50 - 192.168.1.200	192.168.1.0/24 Internal	151 addresses	Wired connection
Laptops (5)	192.168.1.50 - 192.168.1.200	192.168.1.0/24 Internal	151 addresses	Wi-Fi connection
Future devices	192.168.1.201 - 192.168.1.250	192.168.1.0/24 Internal	50 addresses	For expansion
Guest devices	192.168.2.50 - 192.168.2.200	192.168.2.0/24 Guest	151 addresses	Isolated from internal network

IP Address Summary Table

Device Type	Number of Devices	IP Assignment	Subnet
Router / Firewall		1 Static	192.168.1.0/24
Network Switch		1 Static	192.168.1.0/24
NAS Device		1 Static	192.168.1.0/24
Local Server		1 Static	192.168.1.0/24
WAP 1 (Office)		1 Static	192.168.1.0/24
WAP 2 (Guest)		1 Static	192.168.1.0/24
Desktop PCs		20 DHCP (Dynamic)	192.168.1.0/24
Laptops		5 DHCP (Dynamic)	192.168.1.0/24
Guest Devices	Variable	DHCP (Dynamic)	192.168.2.0/24

VLAN Configuration

VLAN ID	VLAN Name	Subnet	Purpose
1	Internal (Default)	192.168.1.0/24	All office devices (desktops, laptops, servers, NAS, printers)
2	Guest	192.168.2.0/24	Guest Wi-Fi - completely isolated from internal network